MSG No. 51.290 $Pmma1'$ [Type II, orthorhombic]

Table 1: Wyckoff site: `2a`, site symmetry: `.2/m.1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,3,5,7,9,11,13,15]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,4,6,8,10,12,14,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `.2/m.1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,3,5,7,9,11,13,15]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,4,6,8,10,12,14,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `.2/m.1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,3,5,7,9,11,13,15]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,4,6,8,10,12,14,16]`

Table 4: Wyckoff site: `2d`, site symmetry: `.2/m.1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,3,5,7,9,11,13,15]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,4,6,8,10,12,14,16]`

Table 5: Wyckoff site: `2e`, site symmetry: `mm21'`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, z\right]$	`[1,4,6,7,9,12,14,15]`
2	$\left[\frac{3}{4}, 0, -z\right]$	`[2,3,5,8,10,11,13,16]`

Table 6: Wyckoff site: `2f`, site symmetry: `mm21'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, z\right]$	`[1,4,6,7,9,12,14,15]`
2	$\left[\frac{3}{4}, \frac{1}{2}, -z\right]$	`[2,3,5,8,10,11,13,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `.2.1'`

No.	position	mapping
1	$[0, y, 0]$	`[1,3,9,11]`
2	$\left[\frac{1}{2}, -y, 0\right]$	`[2,4,10,12]`

continued ...

Table 7

No.	position	mapping
3	$[0, -y, 0]$	`[5,7,13,15]`
4	$\left[\frac{1}{2}, y, 0\right]$	`[6,8,14,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[0, y, \frac{1}{2}\right]$	`[1,3,9,11]`
2	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[2,4,10,12]`
3	$\left[0, -y, \frac{1}{2}\right]$	`[5,7,13,15]`
4	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[6,8,14,16]`

Table 9: Wyckoff site: `4i`, site symmetry: `.m.1'`

No.	position	mapping
1	$[x, 0, z]$	`[1,7,9,15]`
2	$\left[x + \frac{1}{2}, 0, -z\right]$	`[2,8,10,16]`
3	$[-x, 0, -z]$	`[3,5,11,13]`
4	$\left[\frac{1}{2} - x, 0, z\right]$	`[4,6,12,14]`

Table 10: Wyckoff site: `4j`, site symmetry: `.m.1'`

No.	position	mapping
1	$\left[x, \frac{1}{2}, z\right]$	`[1,7,9,15]`
2	$\left[x + \frac{1}{2}, \frac{1}{2}, -z\right]$	`[2,8,10,16]`
3	$\left[-x, \frac{1}{2}, -z\right]$	`[3,5,11,13]`
4	$\left[\frac{1}{2} - x, \frac{1}{2}, z\right]$	`[4,6,12,14]`

Table 11: Wyckoff site: `4k`, site symmetry: `m..1'`

No.	position	mapping
1	$\left[\frac{1}{4}, y, z\right]$	`[1,6,9,14]`
2	$\left[\frac{3}{4}, -y, -z\right]$	`[2,5,10,13]`
3	$\left[\frac{3}{4}, y, -z\right]$	`[3,8,11,16]`
4	$\left[\frac{1}{4}, -y, z\right]$	`[4,7,12,15]`

Table 12: Wyckoff site: `8l`, site symmetry: `11'`

No.	position	mapping
1	$[x, y, z]$	`[1,9]`
2	$\left[x + \frac{1}{2}, -y, -z\right]$	`[2,10]`
3	$[-x, y, -z]$	`[3,11]`

continued ...

Table 12

No.	position	mapping
4	$\left[\frac{1}{2}-x,\,-y,\,z\right]$	[4,12]
5	$[-x,\,-y,\,-z]$	[5,13]
6	$\left[\frac{1}{2}-x,\,y,\,z\right]$	[6,14]
7	$[x,\,-y,\,z]$	[7,15]
8	$\left[x+\frac{1}{2},\,y,\,-z\right]$	[8,16]